

Dynamic Audio Signal Processing for Note Frequency Detection.

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Abstract— This paper presents the development and evaluation of a Dynamic Audio Frequency Recognition and Real-Time Musical Notation Display system.

The accurate recognition of audio frequencies and their translation into musical notation is a crucial challenge in the fields of music technology and audio signal processing. This research introduces a system for Dynamic Audio Frequency Recognition and Real-Time Musical Notation Display, designed to analyze and process audio signals with high precision. Leveraging advanced Digital Signal Processing (DSP) techniques, the system extracts fundamental frequency components and maps them to musical notations such as 'Sa Re Ga Ma' in the Indian classical system and their Western equivalents (e.g., C, D, E).

The proposed system employs machine learning-based classification models to enhance adaptability and robustness, even in complex and noisy environments. This paper addresses limitations in traditional transcription methods, such as their reliance on manual input, susceptibility to errors, and lack of real-time feedback. The developed platform provides an intuitive interface that dynamically visualizes musical notes as they are played or sung, making it a valuable tool for musicians, educators, and learners[9].

Experimental results validate the system's high accuracy in frequency recognition and note mapping, demonstrating its effectiveness in diverse musical contexts. Furthermore, the integration of real-time processing capabilities ensures its utility in live performances, music education, and digital composition tools. Future research aims to extend the system's functionality to support polyphonic inputs and adapt it to global musical systems for broader applicability.

Keywords: Digital Signal Processing (DSP), Audio Frequency Recognition, Musical Notation, Machine Learning, Real-Time Systems.

I.INTRODUCTION

Music has a remarkable ability to connect people across cultures and generations. It can evoke powerful emotions, from joy to sorrow, and everything in between. Music's universal language transcends barriers, making it a unique and cherished form of human expression[4]. Dynamic Audio Frequency Recognition and Real-Time Musical Notation Display involve the process of analyzing audio signals to identify musical frequencies and convert them into written music in real time. This topic is New in the field of audio processing and also very useful to vocalists or instrumentalists for easily getting notations of any song they want. This is a unique project by

which we are giving great help to musicians in the field of music. This project is highly useful for finding interlude pieces, prelude pieces in any song which is most difficult for a musician to find out and reproduce accurately. This critical task is tedious for a person but very easy for a computer process[2].

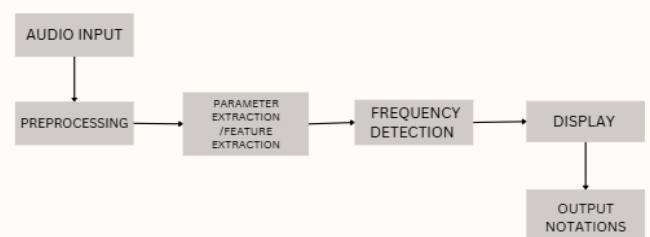
II.LITERATURE SURVEY

Numerous researchers have developed different methods for Dynamic Audio frequency recognition and real-time notation display. The authors have approached, Audio classification for retrieving music information from Hindustani Vocal music, mainly Convolutional Neural Networks (CNN)[1], whereas CNN is a machine learning model widely used to process images for various purposes such as segmentation and classification.[2] The prior methods were primarily concerned with recurrent neural networks (RNNs) for learning complex features in spectrograms or time-series audio data. The music aims to use the vibrato time pattern of Tenuto pitch chords as a change in the arrangement [2]. One of the main problems of using low-frequency sounds for audio analysis. Music serves as a wonderful medium for expressing emotions about the world. The language of music is incredibly diverse. Classical music is one of the main components of Indian societal tradition. The two main subgenres of Indian classical music are Carnatic and Hindustani [3]. Harmonic Product Spectrum (HPS)[4], which mainly focuses on pitch detection, Dynamic Time Warping (DTW), which aids in matching patterns in time-series data, and adaptive signal processing—these are some techniques that researchers worked on which require adjusting the model's parameters to strike a balance between stability and responsiveness [7]. The author worked on research and investigated the Hilbert Spectrum (HS) derived from various AM-FM [6]. The author elaborates on Audio Recognition using Mel Spectrograms and Convolution Neural Networks [8]. It is well-known that CNNs have high performance for image classification. In [1], CNN architecture is performed when used for audio classification.

III.METHODOLOGY

INITIAL WORKPLAN (BLOCK DIAGRAM)

Block Diagram



Despite lots of attempts at single musical "swar"

detection using machine learning algorithms, we faced many challenges, like the model not working properly on a short duration signal that gave an inconsistent result [1]. We investigated anomaly detection methods in light of the machine learning approach's shortcomings. These techniques performed rather well on smaller datasets, but when used on larger datasets, they showed a tendency to overfit. Notwithstanding the encouraging outcomes, the possibility of decreased scalability and generalization led us to look into alternate strategies that could preserve accuracy and robustness over bigger and more varied datasets.

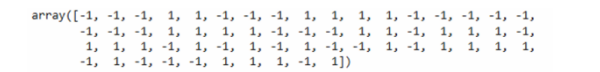


Figure 1: Anomaly Array From Anomaly Detection

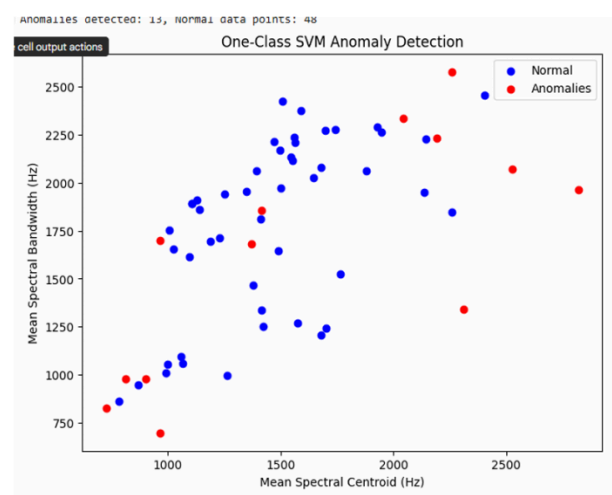


Figure 2: Plot for the same Array

● Input audio acquisition

This system allows users to upload audio files in two formats (i.e., WAV and MP3). After uploading the audio, it starts preprocessing the file. In preprocessing, it trims the silent part of the audio and then samples the file. So the system can process audio properly.

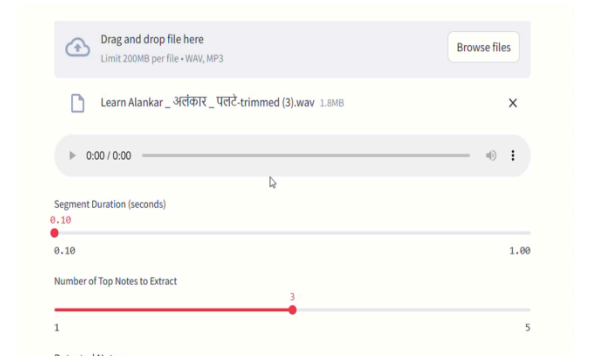


Figure 3: Input as an audio file

● Output

This system is providing 2 types of output

1. Tabular note display: it is a table listing timestamp, detected notes and frequency
 - real-life application: music instructors, beginners and instrumentalists can use this data to create accurate music for given vocal audio signals

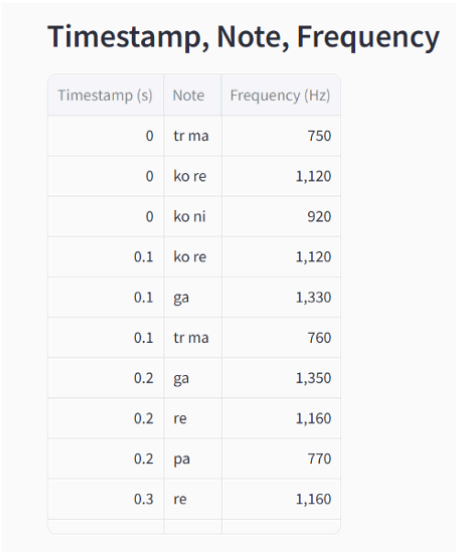


Figure 4: Output Notes

2. Frequency spectrum plot: It is a graphical representation of frequency spectrum for entire audio file where peak corresponds with detected notes.

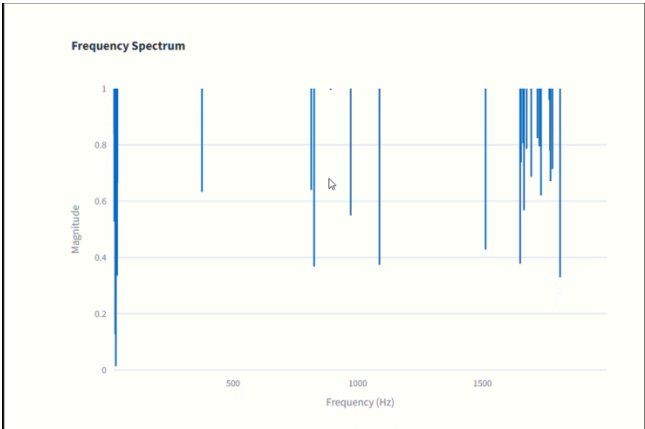


Figure 5: Frequency spectrum output

You can choose how many top notes to extract and change the segment time, among other adjustable options, in the system.

These capabilities guarantee that the system can be customised to accommodate various musical tastes or user preferences. Example optimisation: Depending on the piece's complexity, users can alter the segment duration to detect notes more quickly or in depth.

Along with this we tried some more Machine learning aproch for perfect note detection for that audio file.

Here is the flow of working on that methods:

A. Preprocessing:

The audio data is also preprocessed to make it cleaner and more appropriate for further analysis:

1. Trimming and Normalization — The audio is trimmed to cut silenced or truncate duds and normalized to bring to ensure equal dB spallation ness
2. Alignment: The files are aligned in the time domain to allow for correct syn
3. Prepossessing: Correct manipulation of the data to ensure proper timestamp synchronization, which is vital for correct analysis
4. Good Observations: very few black areas after prepossessing in the spectro grams, suggesting good audio extraction.

B. Feature Extraction:

1. Mean and Standard Deviation:

- Mean: 3.178384 • Standard Deviation: 17.237559

2. Skew and Kurtosis:

- Skew: 48.108430 • Kurtosis: 3449.344531

3. Spectrogram: A graphical representation of the audio signals, displaying that that energy is evenly spread out[8]

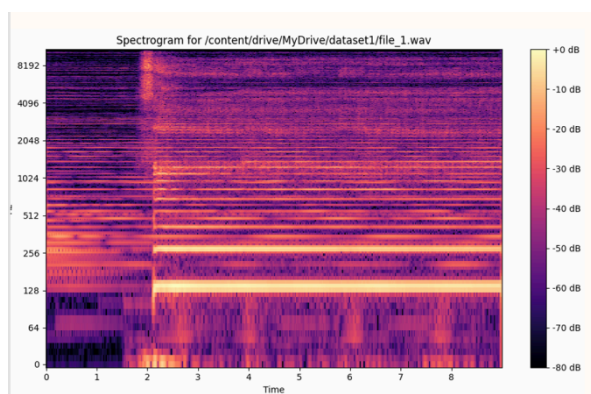


Figure 6: Spectrogram

4. Signal Spectrum: The energy of the signal in frequencies

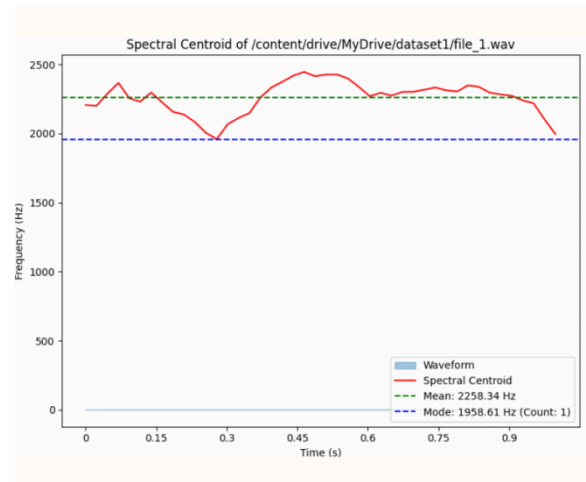


Figure 7: Signal Spectrum

5. Spectral Bandwidth and Flatness: Energy is concentrated with the 2000–2500 Hz frequency range.

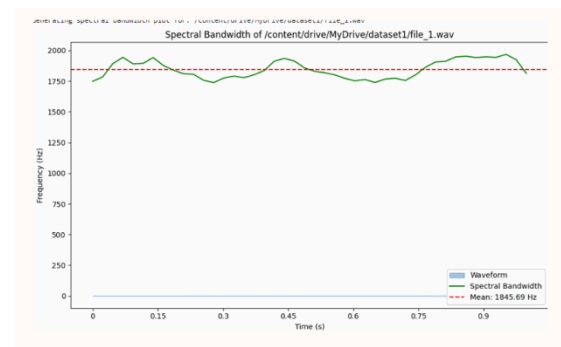


Figure 8: Spectral Bandwidth

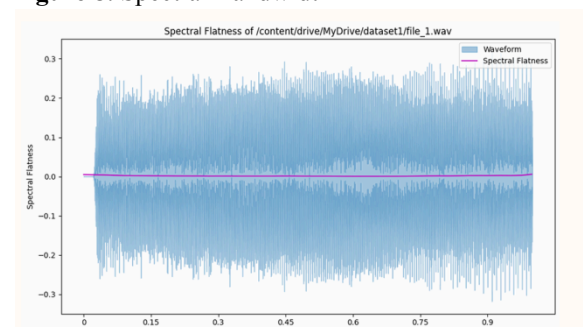


Figure 9: Spectral Flatness

6. Root Mean Square Energy (RMSE): Calculated for each audio file to quantify signal strength.

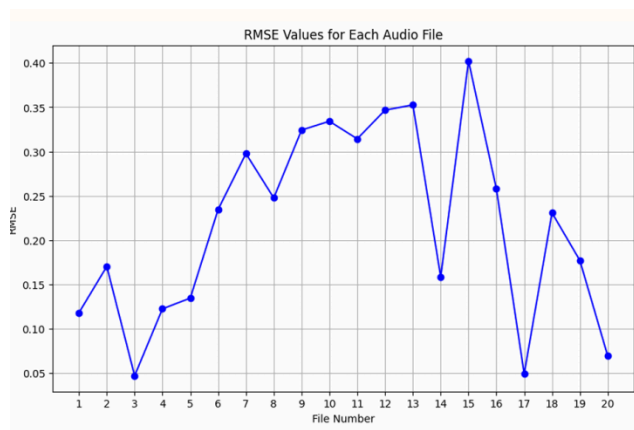


Figure 10: RMSE values

7. Zero-Crossings: Mean value computed as 2354 for the dataset.

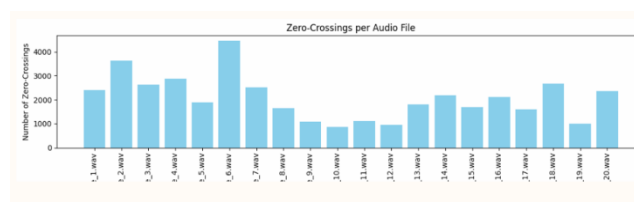


Figure 11: Zero Crossing

8. Mel-Frequency Cepstral Coefficients (MFCC): Extracted for each audio file to represent audio features compactly.

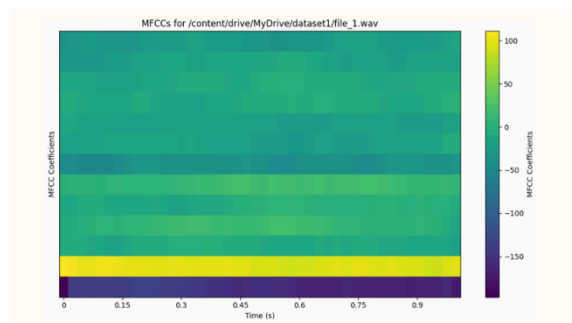


Figure 12: MFCC

Training and Testing with Supervised Learning

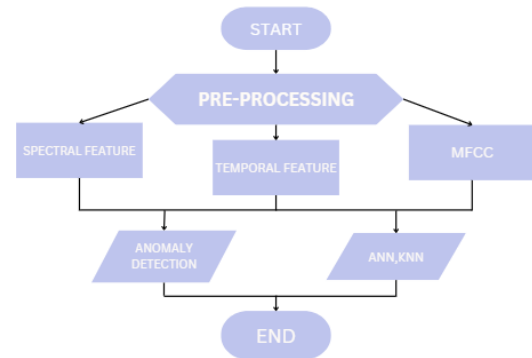
1. Datasets:
 - Training and testing sets split as 20:20 (e.g., (16, 23), (4, 23), (16, 4)).
2. Algorithms:
 - (a) Artificial Neural Network (ANN): 17
 - Test Loss: 0.014790430665016174
 - Test Accuracy: 1.0
 - (b) K-Nearest Neighbour (KNN):
 - Test Accuracy: 1.0

Visualization and Display

- Real-time visualization of musical notations ("Sa," "Re," "Ga," "Ma," "Pa," "Dha," "Ni," "Sa") is performed using Streamlit.

- Audio playback and notation display features allow users to verify the accuracy of the detected notes.

WORKFLOW:



IV. ADVANTAGES, LIMITATIONS, AND CHALLENGE

A. Advantages

1. Real-Time Processing

Converts audio signals into musical notations in real-time, providing instant feedback to users.

Ideal for live performances, practice sessions, and interactive music learning environments.

2. Enhanced Music Education

Assists learners by visually displaying musical notes as they sing or play, bridging the gap between auditory perception and

theoretical understanding.

Simplifies music theory for beginners by dynamically linking sounds to notations.

3. Accuracy and Precision

Utilizes advanced Digital Signal Processing (DSP) and machine learning techniques for accurate frequency recognition, even in noisy environments.

Reduces errors compared to manual transcription methods.

4. Time-Efficient and User-Friendly

Automates the process of transcribing music, saving significant time for musicians and composers.

Features an intuitive interface suitable for users with varying levels of musical expertise.

5. Versatility

Supports multiple musical notation systems, including Indian classical ('Sa Re Ga Ma') and Western notations (e.g., C, D,E).

Can be applied to various musical instruments and vocal inputs.

6. Noise Resilience

Employs noise-filtering algorithms, ensuring effective operation in diverse acoustic environments such as studios, classrooms, or concerts.

7. Applications Across Domains

Useful for music education, live performance analysis, and digital composition[5].

Enables researchers and developers to explore new avenues in music technology and sound analysis.

8. Scalability and Future Expansion

Potential to handle polyphonic inputs (multiple notes played simultaneously) for more complex music.

Can adapt to different musical cultures and systems worldwide, promoting global accessibility.

9. Cost-Effective

Reduces dependency on expensive transcription services or software, making it accessible to a wider audience.

10. Encourages Creativity
Helps musicians experiment with new compositions by providing immediate visual feedback, fostering innovation in music creation.

B. Limitations and Challenges

The system may struggle with accurately recognizing multiple simultaneous notes (polyphony), especially in complex musical compositions.

Noise Sensitivity

Although noise-filtering algorithms are implemented, extreme background noise or overlapping sounds can reduce the accuracy of frequency detection.

Pitch Accuracy

The system may have difficulty detecting microtonal variations or notes that fall between standard musical pitches, which could limit its application for certain music genres.

Instrument-Specific Variability

The system is optimized for general audio signals and may not fully account for the unique tonal qualities or harmonics of specific instruments, leading to inaccuracies.

Real-Time Processing Constraints

High computational requirements may affect the real-time performance of the system on devices with limited processing power, especially in complex or polyphonic scenarios.

Latency Issues[3]

There could be a slight delay between sound input and the corresponding display of musical notations, which may affect live performance applications.

Scalability Concerns

As the system grows in complexity, such as with the addition of more advanced features (e.g., polyphony, advanced musical systems), scalability and performance may become a concern.

Dependency on Clean Audio Input

The accuracy of note detection is heavily reliant on the quality of the input audio. Poor-quality recordings or distorted signals can lead to incorrect or missed notations.

Limited Support for Complex Notations

The system is currently focused on basic notations, and may not support more complex musical symbols (e.g., dynamic markings, articulations) without further development.

Learning Curve for Users

While the system is user-friendly, novice users may require some time to adapt to the visual representation of musical notations, particularly those unfamiliar with music theory.

V.RESULTS

1. supervised learning

Artificial neural network

Test Loss: 0.014790430665016174

Test Accuracy: 1.0

K nearest neighbor KNN

Test Accuracy: 1.0

2. Training and testing sets:20 20 (16, 23) (4, 23) (16,4)

3.Displaying “Sa” “Re” “Ga” “Ma” “Pa” “Dha” “Ni” “Saa”

Previously we done many operations like Noise removing, Feature extraction, Anomaly detection, Training and testing and accuracy of audio files. From this we can display notations of that audio file we inserted as input and it is displaying correct notation of that file as “sa (C4)” or “re (D4)” etc.

⇒ C4 detected!
D4 detected!
E4 detected!
F4 detected!
G4 detected!
A4 detected!
B4 detected!
C5 detected!

Figure 13 : Display of single notes

For this we used YiIN pitch algorithm and MFCC and after that displaying methods.

4.Frequency Recognized output by following methods:

1)FFT (Fast Fourier Transform)[6]:

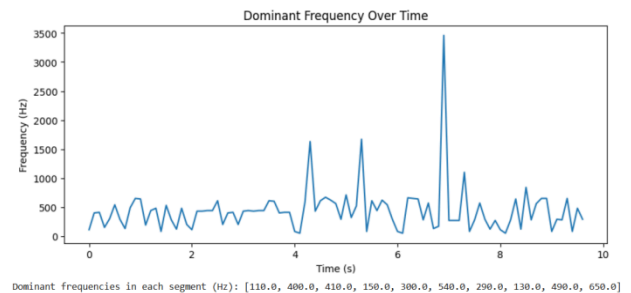


Figure 14: FFT output

2)Hilbert Transform

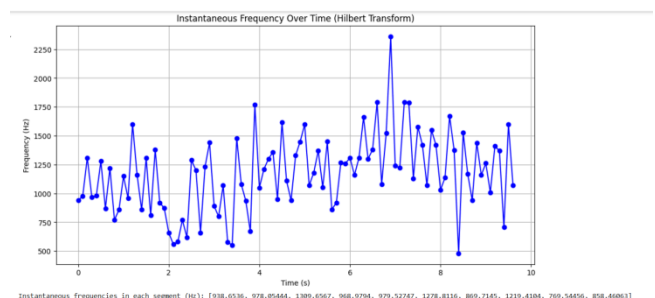


Figure 15: Hilbert output

3)Peak Detection

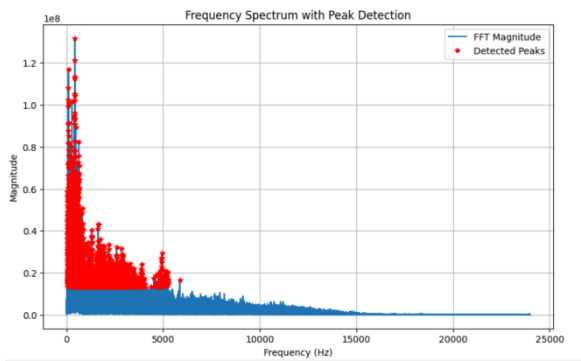


Figure 16:Peak Detection output

4) Cepstrum Analysis[7]

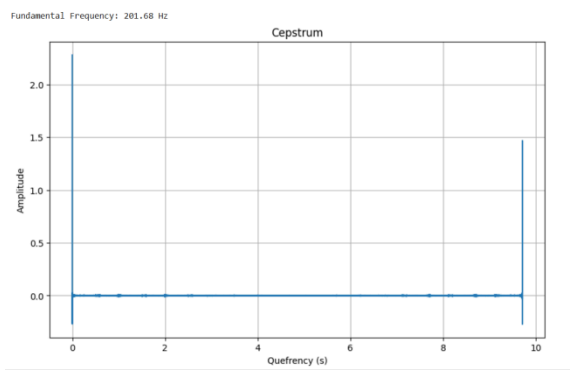
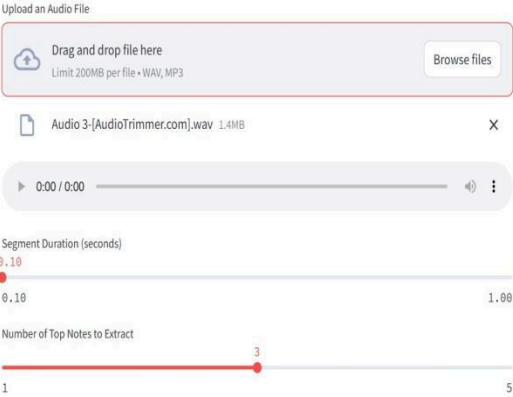


Figure 17: Cepstrum output

5).Display of final output notations using streamlit at different time segment

1) t=0.1sec

Indian Classical Music Note Detection



Timestamp, Note, Frequency

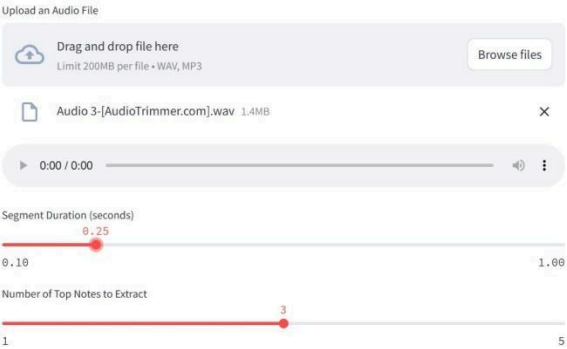
Timestamp (s)	Note	Frequency (Hz)
0	tr ma	744.07
0	ko re	1,120.1
0	re	1,148.1
0.25	ga	1,344.12
0.25	re	1,152.1
0.25	pa	772.07
0.5	re	580.05
0.5	pa	772.07
0.5	ko re	568.05
0.7499	ga	648.06

Timestamp, Note, Frequency

Timestamp (s)	Note	Frequency (Hz)
0	tr ma	750
0	ko re	1,120
0	ko ni	920
0.1	ko re	1,120
0.1	ga	1,330
0.1	tr ma	760
0.2	ga	1,350
0.2	re	1,160
0.2	pa	770
0.3	re	1,160

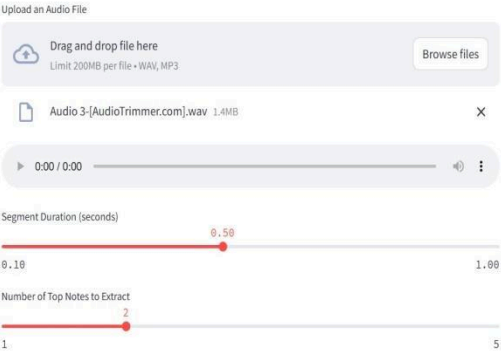
2) t=0.25sec

Indian Classical Music Note Detection



3) $t=0.5\text{sec}$

Indian Classical Music Note Detection



Timestamp, Note, Frequency

Timestamp (s)	Note	Frequency (Hz)
0	re	1,152
0	ga	1,346
0.5	pa	774
0.5	re	580
1	dha	218
1	ga	650
1.5	tr ma	726
1.5	ko ga	1,210
2	ga	1,284
2	pa	770

Comparison Table

Aspect	FFT (Fast Fourier Transform)	Wavelet Transform	Machine Learning (ML)
Purpose	Converts audio signals into frequency components.	Analyzes signals in both time and frequency domains	Learns patterns in audio data to classify frequencies or notes
Time-Frequency Resolution	Limited; good for stationary signals.	Excellent; adaptable for non-stationary signals.	Depends on the dataset and model architecture.
Real-Time Capability	High; computationally efficient	Moderate; computationally intensive	High with optimized models but requires preprocessing
Accuracy	High for simple signals;	High for complex signals but	Very high for polyphonic signals with

	struggles with polyphonic sounds.	depends on parameters.	proper training.
Ease of Implementation	Straightforward with well-established libraries.	Moderate; requires deeper signal processing expertise	Complex; needs annotated datasets and training.
Output	Frequency components (numerical values).	Frequency components with time localization	Direct classification of notes or frequency categories.
Adaptability	Limited; works best for known frequency ranges.	Flexible; adapts well to varying signals	Highly adaptable but requires retraining for new data.
Scalability	Suitable for small-scale applications	Can handle medium-scale tasks.	Highly scalable with sufficient resources
Suitability for Your Project	Good for initial frequency extraction	Suitable for nuanced signal analysis and polyphonic sounds	Best for classification and real-time notation generation

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

The project successfully meets its objectives by providing a system capable of:

- Recognising and detecting audio frequencies of individual musical notes, such as "Sa," "Re," and other classical notations.
 - Displaying these notations in real-time using advanced signal processing techniques.
 - Analysing Alankaras in classical music and mapping them accurately to their corresponding frequencies and notations.
- The results demonstrate that the system effectively processes audio files, extracts relevant features, and translates them into visual musical notations with high accuracy. This indicates that the project achieves its intended goal of aiding classical music analysis and education[9].

B. Future Scope

The project opens several avenues for future work:

- Polyphonic Music Analysis: Extend the system to handle multiple simultaneous notes and more complex compositions.
- Enhanced Machine Learning Models: Incorporate deep learning techniques to improve feature extraction and frequency detection accuracy.
- Robust Noise Handling: Develop advanced noise cancellation algorithms to enhance performance in noisy environments
- Wider Genre Support: Adapt the system for use with different musical traditions and genres beyond classical music.

- Mobile and Web Integration: Create a mobile application or web-based platform for real-time usage and greater accessibility.
- Educational Applications: Expand the system for use in music education, enabling students to learn and practice classical music more effectively.

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